

Geoffrey Barini

Education

- 2015–2018 **Pan African University and Jomo Kenyatta University of Agriculture and Technology**,
Computational Mathematics, Doctor of Philosophy.
- 2010–2012 **Jomo Kenyatta University of Agriculture and Technology**,
Mathematics, Master of Science.
- 2005–2009 **Jomo Kenyatta University of Agriculture and Technology**,
Mathematics, Bachelor of Science.
First Class Honors

Experience

- July 2019–Present **Lecturer/Research Fellow, Jomo Kenyatta University of Agriculture and Technology, Kenya**,
Full Time.
- Deliver undergraduate and postgraduate courses in mathematics, artificial intelligence, and computational modelling.
 - Conduct and publish research in computational mathematics, artificial intelligence, epidemiological modelling, and climate change modelling.
 - Supervise and mentor undergraduate and postgraduate research in mathematics, artificial intelligence, and computational modelling.
- Feb 2026–Present **LLM AI Research Contractor, Turing, USA**, *Freelance (Part-Time, Remote)*.
- Produce high-quality datasets and technical content to support large-language-model training.
 - Perform data annotation, labeling, and editing of technical and instructional content.
 - Review and evaluate AI model outputs to assess accuracy, reasoning quality, and reliability.
 - Support software tasks and conduct quality-assurance activities to improve model performance and training workflows.
- Jan 2013–June 2019 **Tutorial Fellow, Jomo Kenyatta University of Agriculture and Technology, Kenya**, *Full Time*.
- Deliver tutorials and assist in teaching undergraduate mathematics and artificial intelligence courses.
 - Prepare teaching materials, assignments, and assessments to support course delivery.
 - Provide academic support and basic supervision for undergraduate projects and coursework.
- March 2010–Dec 2012 **Teaching Assistant, Jomo Kenyatta University of Agriculture and Technology, Kenya**, *Full Time*.
- Support lecturers by conducting tutorials and helping students understand course materials.
 - Assist in grading assignments, preparing teaching materials, and organizing course resources.
 - Provide guidance to undergraduate students during coursework and practical sessions.

Selected Publications

- **Geoffrey Barini** and Walter Mogeni and Mogeni Polycarp, *Mental health outcomes and ambient environmental exposures in Mozambique: a population-based nationally representative survey*, Submitted, 2025, <https://www.medrxiv.org/content/10.1101/2025.10.29.25339102v1>.
- Linnet Onger, Lydia Kaduka, Damaris Matoke, Doris Njomo, Zipporah Bukania, Moses Mwangi, Linus Ndegwa, Joanna Olale, Caleb Othieno, Sahara Hussein, **Geoffrey Barini**, Valentine Wanga and Polycarp Mogeni, *Socioeconomic inequalities, substance use, and chronic multimorbidity in Kiambu and Nakuru counties in Kenya*, **Plos Global Public Health**, 2025, <https://doi.org/10.1371/journal.pgph.0005544>.
- **Geoffrey Barini** and Onger Linnet and Mogeni Polycarp, *Mental Health Outcomes and Intimate Partner Violence among Nepalese Women: A Propensity Score Matched Study*, **Plos Mental Health**, 2025, <https://doi.org/10.1371/journal.pmen.0000374>.
- **Geoffrey Barini**, Amima Sharon, Mulwa Damaris and Mogeni Polycarp, *Spatial heterogeneity of menstrual discriminatory practices against Nepalese women: a population-based study using the 2022 Demographic and Health Survey*, **Plos Global Public Health**, 2024,

<https://doi.org/10.1371/journal.pgph.0003145>.

- **Barini Geoffrey**, Ngoo Livingstone and Mwangi Ronald, *Application of a Fuzzy Unit Hypercube in Cardiovascular Risk Classification*, **Soft Computing**, 2019, <https://doi.org/10.1007/s00500-019-03802-0>.
- **Barini Geoffrey**, Ngoo Livingstone and Mwangi Ronald, *Some new fuzzy entropy formulae*, **International Journal of Computing and Optimization**, 2018, <https://doi.org/10.12988/ijco.2018.846>.
- **Barini Geoffrey**, Ngoo Livingstone and Mwangi Ronald, *Measuring Fuzzy Specificity Using Fuzzy Unit Hypercube*, **International Journal of Computing and Optimization**, 2018, <https://doi.org/10.12988/ijco.2018.844>.

Active Research

Nov 2025– **Geoffrey Barini and Polycarp Mogeni**, *Ambient temperature and intake of sugar sweetened beverages among women across 12 Sub-Saharan Africa countries: a cross sectional analysis*.

Peer-Review Activities

- BMJ Global Health (British Medical Journal), 2025–Present
- Global Mental Health (Cambridge Prisms), 2026–Present

Administrative Appointments

2026–Present **External Examiner**,
Mathematics, African Nazarene University, Nairobi, Kenya.

2014–Present **Academic Programmes Coordinator**,
Department of Pure and Applied Mathematics,
Jomo Kenyatta University of Agriculture and Technology.

2019 **Acting Chairman of Department**,
Department of Pure and Applied Mathematics,
Jomo Kenyatta University of Agriculture and Technology, Nairobi, Kenya.

2014 **Examination Officer**,
Department of Pure and Applied Mathematics,
Jomo Kenyatta University of Agriculture and Technology, Nairobi, Kenya.

Skills

Languages Python, R

GIS and Remote Sensing QGIS, ArcGIS, Google Earth Engine(GEE)

Big Data and Databases Apache Spark, MySQL, MongoDB, dtb

Frameworks Keras, Pandas, Numpy, PyTorch, Tensorflow, Dash and R shiny

WebD HTML/CSS, JavaScript

Referees

Dr. Polycarp Mogeni,
Epidemiologist,
Kenya Medical Research Institute, Nairobi, Kenya,
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Prof. Livingstone Ngoo,
DVC Academic, Affairs,
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Dr. Lewis Nyaga,
*Chairman, Department of Pure and
Applied Mathematics, JKUAT,*
Nairobi, Kenya,
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